

UNIVERSITY OF TECHNOLOGY - VNUHCM	HOMEWORK 2	Semester/Academic year	1 / 2026-2027
FACULTY OF COMPUTER SCIENCE AND ENGINEERING	Decision Trees: Impurity, Splits, Pruning, and Robust Use	Due	07/10/2026 07:00
	CO3117 - Machine Learning	Format	Formative take-home

Purpose. Understand how a tree chooses and controls splits without reproducing any midterm question pattern or numerical dataset.

- Complete the work by hand where calculation or derivation is requested. Show reasoning; unsupported final numbers receive limited credit.
- Use lecture notes and course resources. Do not use AI, online solution lookup, or classmates' answers for your first attempt. Later supportive AI use must follow the course AI_USE.md protocol.
- These activities are designed for concept transfer and assignment support. They intentionally do not reproduce the wording, ordering, numerical data, or question structure of the midterm.
- No official worked solutions will be released. Students are encouraged to self-check their handwriting with course materials, discussion, and AI assistants after completion.
- Submit one PDF in your course repository and preserve your first attempt. Corrections may be added later without replacing the original submission.

Activity 1 - Impurity intuition before calculation (1.5 pts)

For three binary-class nodes with counts (12,0), (9,3), and (6,6), rank them from lowest to highest impurity. Justify the ranking using entropy intuition, then verify by computing entropy for the three nodes.

Activity 2 - Compare two candidate splits with Gini (2.0 pts)

- A parent node has 18 samples: 11 Positive and 7 Negative. Split S1 produces children (6P,1N) and (5P,6N). Split S2 produces children (8P,4N) and (3P,3N). Compute the weighted Gini impurity for both splits and state which split CART would prefer.
- Explain why the preferred split is not necessarily the split that creates the most balanced child sizes.

Activity 3 - Continuous attributes as a search problem (2.0 pts)

- You observe one numeric feature with values 4, 7, 9, 13, 18, 21, 29 together with binary labels of your choice. Construct three plausible cut points, compute a simple impurity score for each, and explain how a tree-search procedure would compare them.
- Your chosen labels must make at least two candidate cut points genuinely different in quality.

Activity 4 - Pruning and missing values (2.0 pts)

- A subtree makes 7 errors on 30 validation examples. Replacing it with a leaf makes 6 errors. Under reduced-error pruning, what should be done and why?
- Give two sensible strategies for handling a missing value at a split and explain one drawback of each.

Activity 5 - Assignment bridge: tree audit (2.5 pts)

- Run or inspect one decision-tree model on your longitudinal-assignment dataset. Record: maximum depth, number of leaves, training score, validation score, and the top three features used by the tree (or explain why such a ranking is unavailable).
- Change one complexity-control setting (e.g., max depth, min samples, pruning parameter). Predict the effect before running it, then compare the actual result with your prediction in 5-8 sentences.

Submission checklist

- ☐ Name and student ID on first page
- ☐ All calculations show intermediate steps
- ☐ Assignment-bridge activity uses your own dataset/use case
- ☐ First attempt preserved
- ☐ Any later AI/reference use disclosed according to course policy